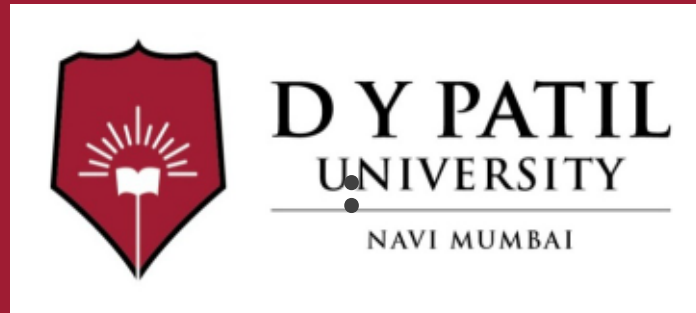




Statistical Techniques

Unit VI : Dimensionality Reduction Techniques (PCA & LDA)

Faculty Name :Dr. Ashwini Kadam

Syllabus

- **Introduction to Dimensionality Reduction:** Concept of dimensionality reduction, Need and benefits of dimensionality reduction, PCA and LDA



Introduction to Dimensionality Reduction

Dimensionality Reduction is a technique used in Machine Learning and Data Science to reduce the number of input variables (features) in a dataset while preserving important information.

In simple terms:

It means converting large, complex data (many columns) into a smaller set of meaningful features.



Example

Suppose a student dataset has 10 features:

Marks in 5 subjects

Attendance

Participation

Assignment score

Project score

Internal assessment

Instead of using all 6, we can reduce them into:

Academic Performance

Engagement Score

So, 6 variables → 2 meaningful variables



Need for Dimensionality Reduction

Dimensionality reduction is required due to the following problems:

1. Curse of Dimensionality

As the number of features increases, data becomes sparse and difficult to analyze.

Example:

An e-commerce company tracks:

100 customer features (age, clicks, time spent, device, etc.)

With too many variables:

Data points become scattered

Patterns are hard to detect

→ Reducing features helps in identifying meaningful customer behavior.



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2. High Computational Cost

More features require more processing time and power.

Example:

A machine learning model using:

200 features → takes 10 minutes to train

20 features → takes 1 minute

→ Reducing dimensions speeds up computation significantly.

Need for Dimensionality Reduction

3. Overfitting Problem

Too many features may capture noise instead of actual patterns.

Example:

A student performance model includes:

Marks, attendance, plus irrelevant features like favourite colour

Model may learn unnecessary patterns

➡ Removing irrelevant features improves prediction accuracy

4. Difficulty in Data Visualization

Humans can visualize only 2D or 3D data effectively.

Example:

A dataset has:

8 variables

Cannot plot directly

→ Reduce to 2 variables → can plot graph and identify clusters easily

Need for Dimensionality Reduction

5. Redundant and Irrelevant Data

Some features may provide duplicate or unnecessary information.

Example:

A dataset includes:

Total marks

Percentage

Both convey similar information

→ One can be removed without losing meaning



6. Storage Issues

Large datasets require more memory.

Example:

An image dataset:

Each image = 1000 features

After reduction:

Only 100 features needed

 Saves storage and improves efficiency

7. Improves Model Interpretability

Too many variables make models complex and hard to understand.

Example:

A business model uses:

50 variables

Difficult for managers to interpret

→ Reduced to 5 key factors → easier decision-making

Conclusion

Dimensionality reduction is needed to reduce complexity, improve efficiency, avoid overfitting, and make data easier to visualize and interpret.



Benefits of Dimensionality Reduction

Dimensionality reduction helps in simplifying data while retaining important information. It provides several practical advantages in Machine Learning and Data Science.



Benefits of Dimensionality Reduction

1. Improves Model Performance

Reducing irrelevant and noisy features helps models learn better patterns.

Example:

A student performance dataset includes:

Marks, attendance, study hours (useful)

Favourite colour, hobby (irrelevant)

Removing irrelevant features improves prediction accuracy.



2. Faster Computation

Less number of features reduces training and processing time.

Example:

Model with 100 features → takes 15 minutes

Model with 20 features → takes 2 minutes

→ Dimensionality reduction speeds up algorithms significantly.

Benefits of Dimensionality Reduction

3. Reduces Overfitting

Fewer features reduce the chance of the model learning noise.

Example:

In a sales prediction model:

Too many variables may capture random fluctuations

Removing extra features leads to better generalization on new data.

Example:

When dataset has too many features (columns), the model may start learning:

- unwanted patterns
- random mistakes
- noise in data

instead of learning the real relationship.



4. Better Data Visualization

Helps in visualizing high-dimensional data in 2D or 3D.

Example:

Customer data has 10 features

→ Reduced to 2 features → plotted on graph to identify customer segments easily

5. Removes Multicollinearity

Highly correlated variables can affect model performance.

Example:

Dataset includes:

Total marks

Percentage

Both are related → one can be removed to avoid redundancy

6. Saves Storage Space

Less data means less memory requirement.

Example:

Image dataset:

Original = 1000 pixels/features

Reduced = 100 features

→ Saves storage and improves efficiency

7. Simplifies Models

Models become easier to understand and interpret.

Example:

A business model with:

50 variables → complex

After reduction:

5 key factors → easy for managers to understand and use

8. Improves Data Quality

Removes noise and irrelevant information.

Example:

Sensor data may contain:

Errors or unnecessary readings

Dimensionality reduction filters useful information only

9. Enhances Decision Making

Simplified data helps in better and faster decisions.

Example:

Marketing team uses:

3 key customer metrics instead of 20

→ Makes targeting and strategy easier

Conclusion

Dimensionality reduction improves model efficiency, reduces complexity, speeds up computation, and enhances decision-making by removing unnecessary and redundant data.



Approaches to Dimensionality Reduction

Dimensionality reduction can be done using feature extraction techniques, where new variables are created from existing ones. Two important approaches are:

- Principal Component Analysis
- Linear Discriminant Analysis

These are two famous Dimensionality Reduction Techniques used in:

- Machine Learning
- Data Science
- Artificial Intelligence



1. Principal Component Analysis (PCA)

Concept

PCA is an **unsupervised technique** used to reduce dimensionality by transforming original variables into a new set of variables called principal components.

These components:

- Are uncorrelated
- Capture maximum variance in data
- Are ordered by importance

PCA reduces data by finding the most important patterns in the dataset.

Example

Suppose teacher checks student performance using:

- Maths marks
- Physics marks
- Chemistry marks

These subjects are related.

Instead of studying all separately,
teacher creates:

“Science Performance Score”



Linear Discriminant Analysis (LDA)

Introduction

Linear Discriminant Analysis (LDA) is a popular **dimensionality reduction** and **classification** technique used in statistics, machine learning, and data science.

It helps:

reduce the number of variables

separate different classes/groups clearly

Definition

“Linear Discriminant Analysis (LDA) is a technique used to reduce dimensions while maximizing the separation between different classes of data.”



LDA tries to:

Reduce unnecessary features

Keep important information

Make different categories easy to distinguish

Why LDA is Needed

Large datasets may contain:

- too many features
- duplicate information
- noisy data

Problems:

- slower processing
- difficult analysis
- poor prediction accuracy
- LDA solves these problems.



Example- LDA

Email Classification System

An AI system classifies emails into:

Spam

Not Spam

Features used:

Number of links

Repeated keywords

Sender type

Attachments

LDA tries to:

Create dimensions that clearly separate spam and non-spam emails.

Purpose:

Improve classification accuracy.



PCA Analogy

Imagine compressing a large textbook into short notes while keeping important concepts.

→ Main aim = retain important information.

LDA Analogy

Imagine arranging students of different divisions in separate rows so they are clearly distinguishable.

→ Main aim = separate categories/classes.



Difference Between PCA and LDA

Basis	PCA (Principal Component Analysis)	LDA (Linear Discriminant Analysis)
Meaning	Reduces dimensions by finding directions with maximum variance	Reduces dimensions by maximizing separation between classes
Type	Unsupervised learning technique	Supervised learning technique
Uses Class Labels?	No	Yes
Main Objective	Preserve maximum information/variance	Improve class separability
Focus	Captures overall data variation	Separates categories/classes clearly
Best Used For	Data compression, visualization	Classification problems
Output	Principal Components	Linear Discriminants
Accuracy Focus	Does not consider class accuracy	Improves classification accuracy
Example Area	Image compression, recommendation systems	Spam detection, face recognition



Thank You



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